Presentazione Esame Architetture Dati

Dataset Degradation

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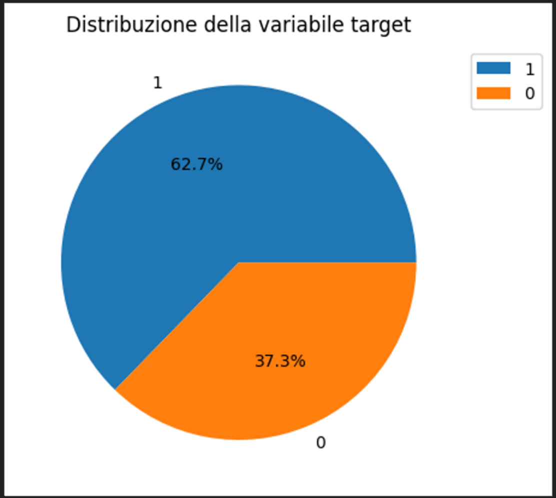
Obiettivo

Questo progetto punta a:

1. Analizzare il Dataset per la diagnosi del carcinoma mammario Wisconsin Breast Cancer Dataset;
2. Introdurre del Data Poisoning all’interno del Dataset;
3. Addestrare 3 modelli differenti con vari livelli di corruzione del Dataset;
4. Osservare come i modelli si comportano al variare della corruzione.

Informazioni sul dataset

Wisconsin Breast Cancer Dataset, di dimensione ridotta, presenta features derivate da osservazioni di masse tumorali, con la rispettiva diagnosi associata.




```
RangeIndex: 569 entries, 0 to 568
Data columns (total 32 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   id                       569 non-null    int64
 1   diagnosis                569 non-null    object
 2   radius_mean              569 non-null    float64
 3   texture_mean             569 non-null    float64
 4   perimeter_mean           569 non-null    float64
 5   area_mean                569 non-null    float64
 6   smoothness_mean          569 non-null    float64
 7   compactness_mean         569 non-null    float64
 8   concavity_mean           569 non-null    float64
 9   concave points_mean      569 non-null    float64
 10  symmetry_mean            569 non-null    float64
 11  fractal_dimension_mean   569 non-null    float64
 12  radius_se                569 non-null    float64
 13  texture_se               569 non-null    float64
 14  perimeter_se             569 non-null    float64
 15  area_se                  569 non-null    float64
 16  smoothness_se            569 non-null    float64
 17  compactness_se           569 non-null    float64
 18  concavity_se             569 non-null    float64
 19  concave points_se        569 non-null    float64
 20  symmetry_se              569 non-null    float64
 21  fractal_dimension_se     569 non-null    float64
 22  radius_worst             569 non-null    float64
 23  texture_worst            569 non-null    float64
 24  perimeter_worst          569 non-null    float64
 25  area_worst               569 non-null    float64
 26  smoothness_worst         569 non-null    float64
 27  compactness_worst        569 non-null    float64
 28  concavity_worst          569 non-null    float64
 29  concave points_worst     569 non-null    float64
 30  symmetry_worst           569 non-null    float64
 31  fractal_dimension_worst  569 non-null    float64
dtypes: float64(30), int64(1), object(1)
```

Data engineering

La colonna “id” è stata rimossa data l’unicità dei valori.

La colonna “diagnosis” è stata trasformata in numerica (1 = maligno, 0 = benigno).

```
df['diagnosis'] = LabelEncoder().fit_transform(df['diagnosis'])
```

```
def standardize(X_tr, X_te):
    """Fit scaler su train, trasforma entrambi; restituisce DataFrame."""    "trasforma":
    scaler = StandardScaler()
    def to_df(arr, ref): return pd.DataFrame(arr, columns=ref.columns, index=ref.index)
    return to_df(scaler.fit_transform(X_tr), X_tr), to_df(scaler.transform(X_te), X_te)
```

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Data columns (total 32 columns):
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 8   concavity_mean
 9   concave points_mean
 10  symmetry_mean
 11  fractal_dimension_m
 12  radius_se
 13  texture_se
 14  perimeter_se
 15  area_se
 16  smoothness_se
 17  compactness_se
 18  concavity_se
 19  concave points_se
 20  symmetry_se
 21  fractal_dimension_s
 22  radius_worst
 23  texture_worst
 24  perimeter_worst
 25  area_worst
 26  smoothness_worst
 27  compactness_worst
 28  concavity_worst
 29  concave points_wors
 30  symmetry_worst
 31  fractal_dimension_w
dtypes: float64(30), int
```

```
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   diagnosis                569 non-null    int64
 1   radius_mean              569 non-null    float64
 2   texture_mean             569 non-null    float64
 3   perimeter_mean           569 non-null    float64
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 5   smoothness_mean          569 non-null    float64
 6   compactness_mean         569 non-null    float64
 7   concavity_mean           569 non-null    float64
 8   concave points_mean      569 non-null    float64
 9   symmetry_mean            569 non-null    float64
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 25  smoothness_worst         569 non-null    float64
 26  compactness_worst        569 non-null    float64
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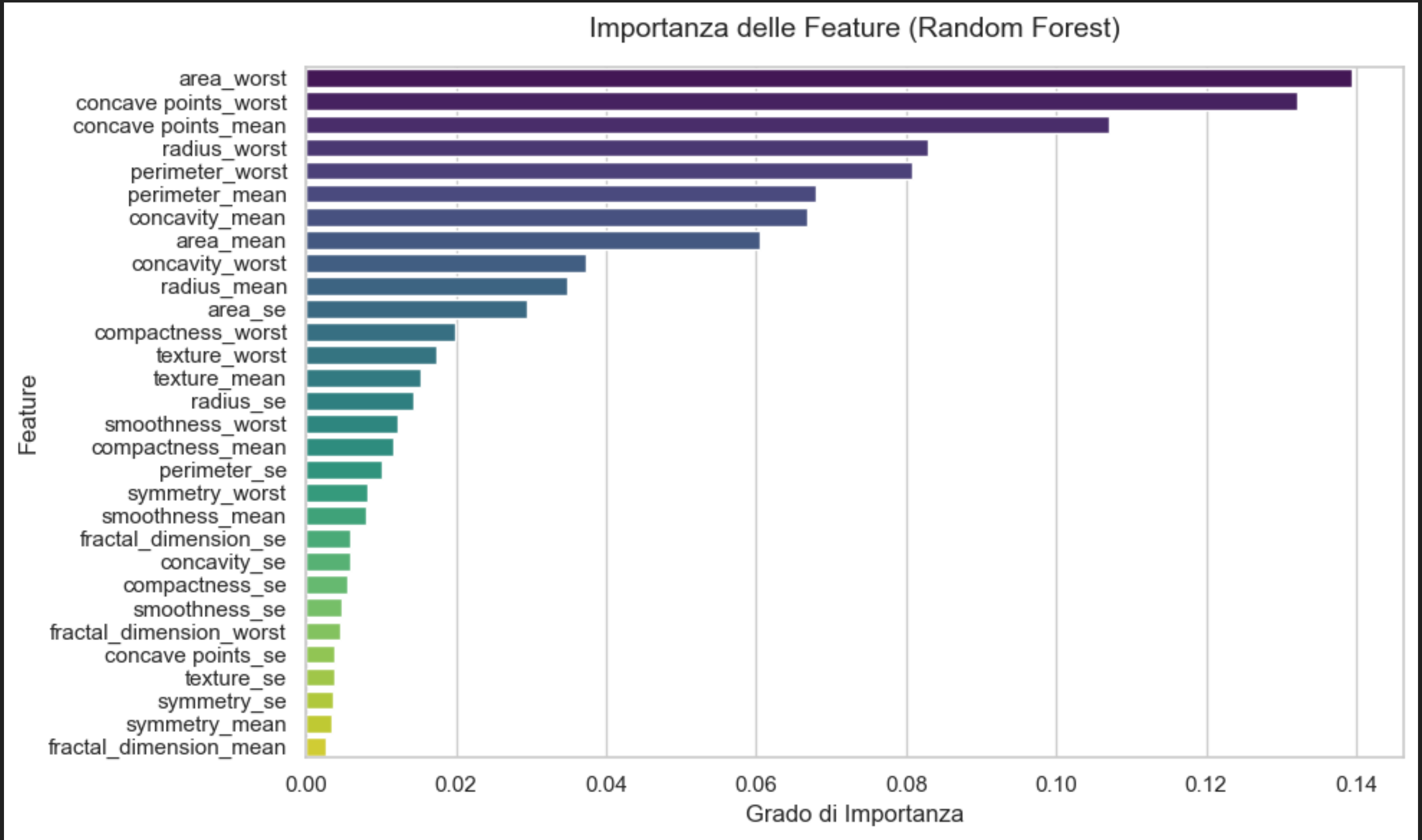
 29  symmetry_worst           569 non-null    float64
 30  fractal_dimension_worst  569 non-null    float64
```

Importanza Feature



Rumori analizzati

- Label Flipping
- Gaussian Noise
- Outlier Injection
- Missing Features
- Data Drift
- Random Feature Drop
- Top-K Drop
- Duplicates

Implementati tramite libreria Pucktrick.

Selezione modelli

Valutando tutti i modelli abbiamo scelto di utilizzare le Reti Neurali (MLP), le Support Vector Machines e la Random Forest.

Le MLP e le Support Vector Machines riescono a gestire bene i valori numerici, particolarmente per la natura binaria del risultato.

La Random Forest è stata scelta per la sua capacità di resistere al rumore.

```
MODELS = {
    'SVM': SVC(kernel='linear', C=1.0, random_state=current_seed),
    'RandomForest': RandomForestClassifier(n_estimators=100, random_state=current_seed),
    'MLP': MLPClassifier(hidden_layer_sizes=(64, 32), activation='relu',
                         alpha=0.0001, early_stopping=True,
                         validation_fraction=0.1, max_iter=1000,
                         random_state=current_seed),
}
```

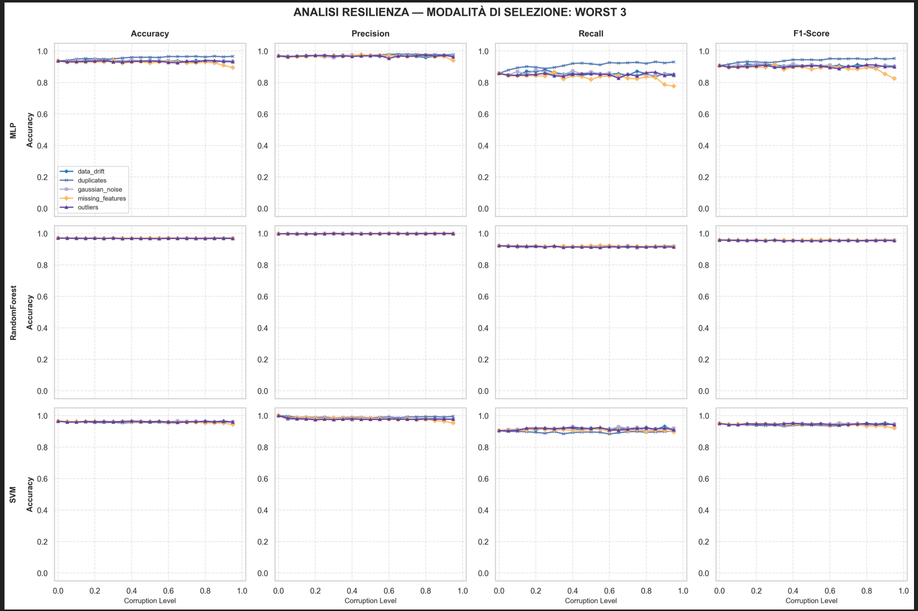
Pipelines

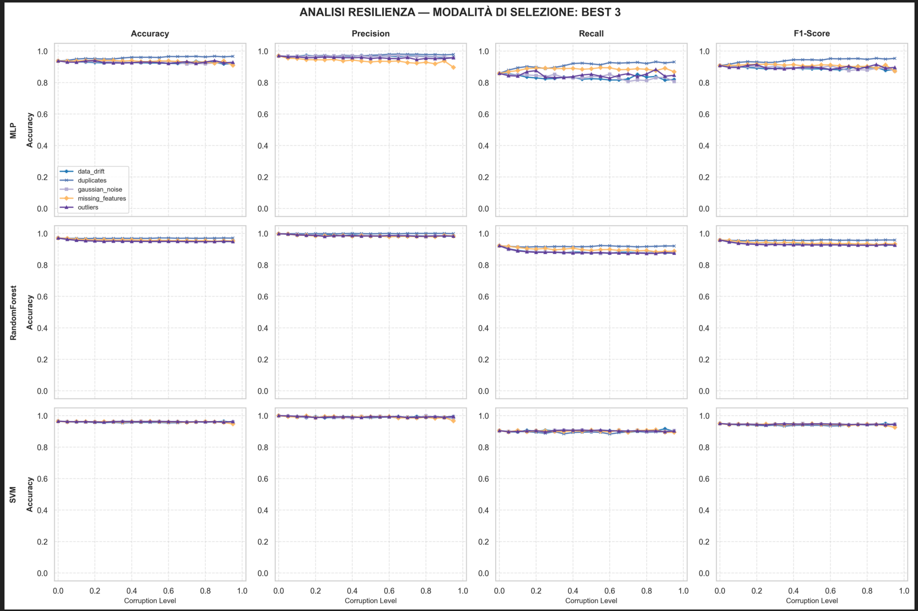
La pipeline sperimentale adottata introduce una struttura a tre livelli annidati rispetto alla baseline originale.

1. Il primo livello riguarda la modalità di selezione delle feature (all features, best 3 features, worst 3 features);
2. Il secondo livello copre il tipo di corruzione;
3. Il terzo livello definisce l'intensità di corruzione (19 livelli da 0.05 a 0.95 con step 0.05, più la baseline pulita a livello 0);
4. Ogni casistica iterata 20 volte per stabilità.

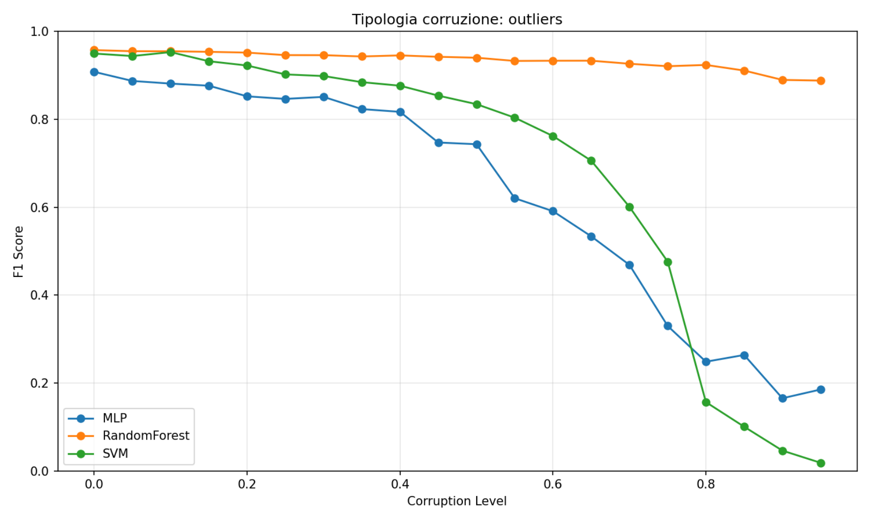
TOTALE: $3 \times 8 \times 20 \times 3 \times 20 = 28.800$ addestramenti.

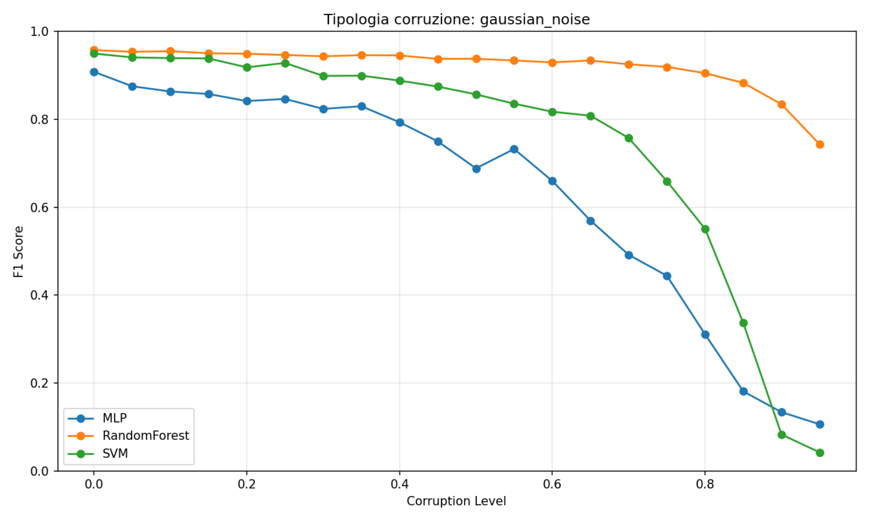
Risultati Worst e Best 3

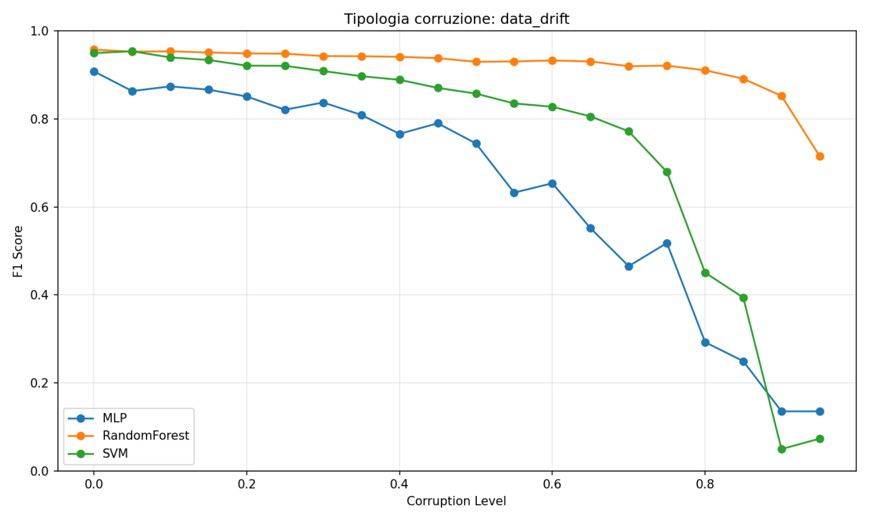




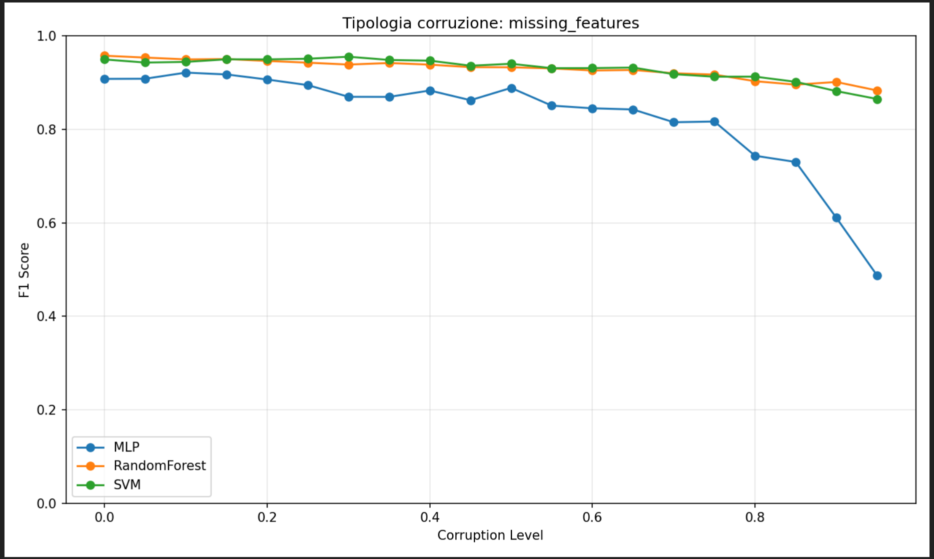
Risultati rumori numerici

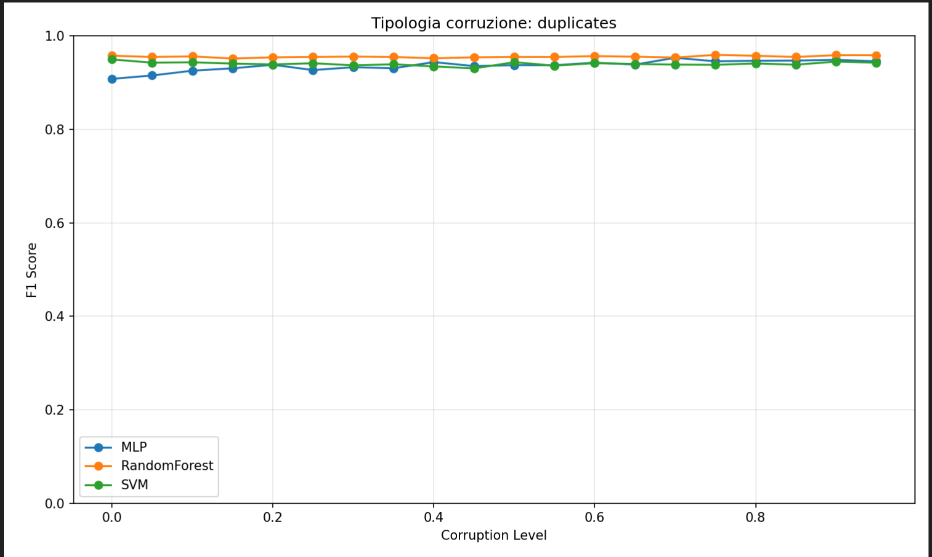




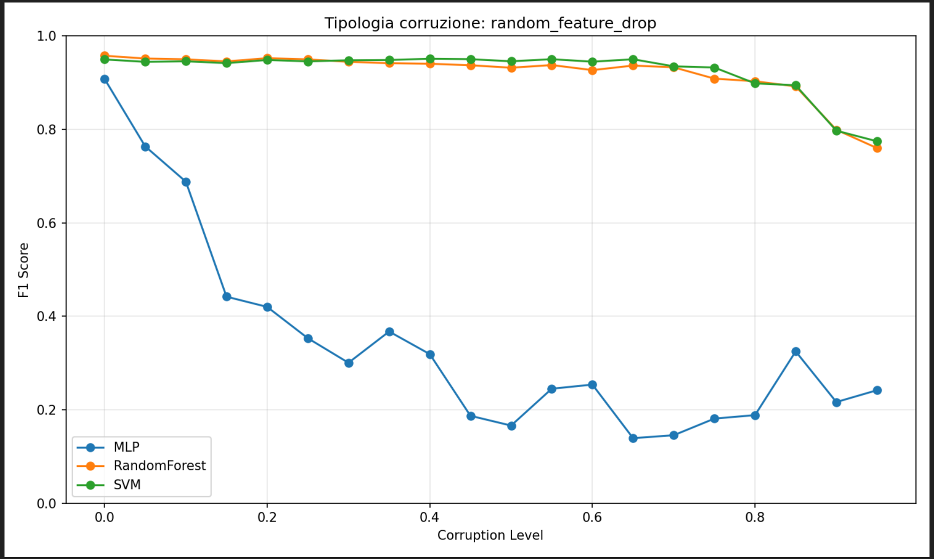


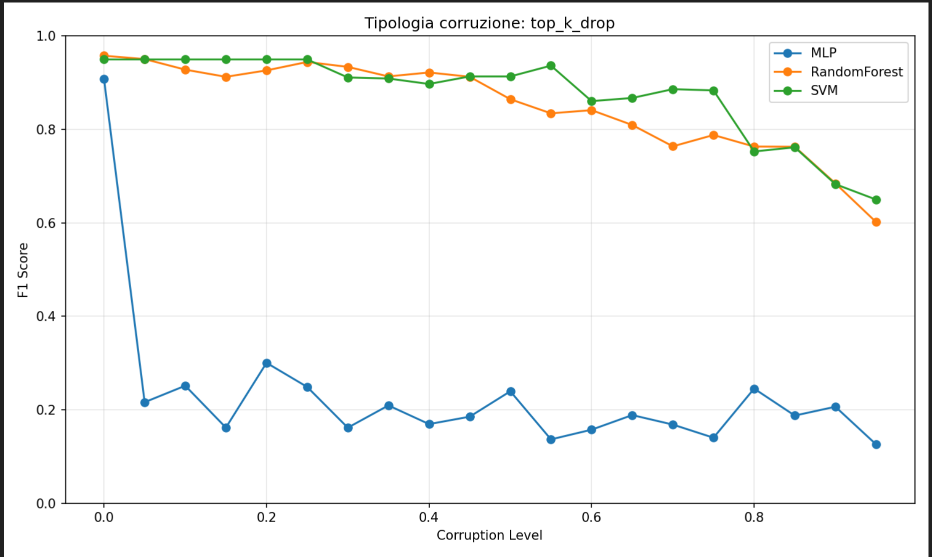
Risultati Missing e Duplicati



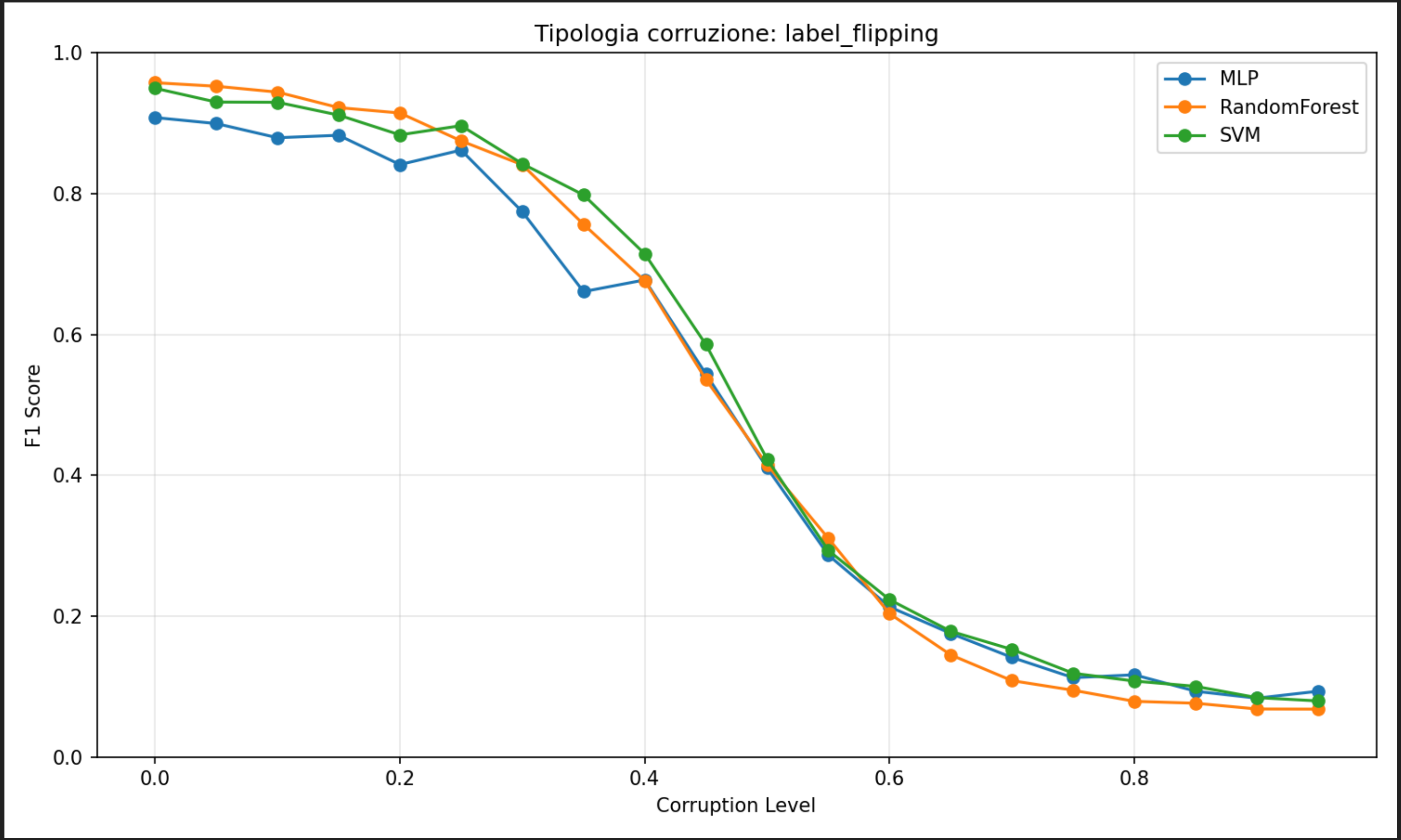


Risultati Drop Feature





Risultati Label Flipping



Conclusioni

- Random Forest è più resistente a interferenze
- La Rete Neurale (MLP) è più prona a degradazione
- Label Flipping crea problemi ovunque
- Duplicates aiuta i modelli andando a migliorare leggermente
- Le Top Feature non sono essenziali se non per MLP

Fine